Scenario + Goal



Network · Demand
SUMO seed

Baseline Diagnosis

Baseline KPIs

Frozen Top-10 + neighbors

Local subgraph

Grounded Initialization

5 shared initial actions

Validate → Modify → SUMO

Measured initial plans

Catalog-Based Refinement

Signal expert

Network expert

Routing expert

2 measured parents

Legal mutation catalog

LLM / Random / Rule

Up to 2 distinct children per parent

Validate → Modify → SUMO

Measured singles

Compatible Composition

Top-4 measured singles

Up to 2 bundles

Joint SUMO

Joint-Feedback Refinement

Measured joint KPIs

Modify one component
Keep partner fixed

Re-simulate joint plan

2 rounds · Beam width 2
Joint children become parents

Measured Candidate Pool

Baseline

Singles

Bundles

Joint refinements

Constrained Selection

- Top-10 gain ≥ 1%
- Global delay increase ≤ 1%
- Neighbor loss increase ≤ 5%
- Throughput ≥ baseline
- Teleports ≤ baseline

Selected Plan + KPI Report



Baseline if no feasible improvement